THIRD EYE NAVIGATION

Project by:

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PROJECT PROPOSAL:

What is **Third Eye Navigation**?

- A wearable device to help people who are blind or visually impaired avoid obstacles while navigating.
- The device will be equipped with an ultrasonic sensor to detect objects in range of the user, and a combination of a buzzer and a vibration motor will be used as the feedback system.
- The device is also Bluetooth compatible, so that users may choose to connect headphones for additional directional cues

HARDWARE DETAILS:

Component	Specific HW	Purpose
Main Processor	ESP32 Microcontroller	Process sensor data & control other modules
Sensor	Ultrasonic/ LiDAR	Object detection
Buzzer	Passive buzzer module	Audio feedback (beeping)
Vibration Motor	--	Tactile feedback (vibration)
Charging Board	--	To charge the device battery
Rechargeable Battery	USB-C chargeable lithium battery module	Power supply
Breadboard	--	To connect modules to ESP32

BACKGROUND:

- Navigation can be difficult and sometimes dangerous for those who cannot rely on their sense of sight to get from place to place
- There are many techniques used by the visually impaired to combat these difficulties, including using a mobility cane, a guide dog, or a human guide to aid in navigation.
- Third Eye aims to be a valuable tool for people who are visually impaired to help them maintain their independence when navigating unfamiliar places.
- Unlike a mobility cane, Third Eye will be able to detect objects not only on the ground, but also obstacles at chest/eye level and objects in motion.

HOW THIRD EYE IS DIFFERENT

- *Walking Sticks*
  - A walking stick requires strong cane skills, relies on how person taps it, can miss and bump into things or someone can just get tired of it and not do it. Human error is a factor. Only 2-8% of visually impaired people use them.
- *Seeing-Eye Dogs*
  - Seeing-eye dogs are expensive; they require food, care, and maintenance(vet visits). Grooming requires them to be without their "eyes" for hours at a time and they retire after 6-8 years. Human interference can mess up the dog's work.
- *Why we're different*
  - Less room for human error, will continuously send out signals, has a larger range to detect obstacles other methods might miss.

DIAGRAM 1:
USER MODEL

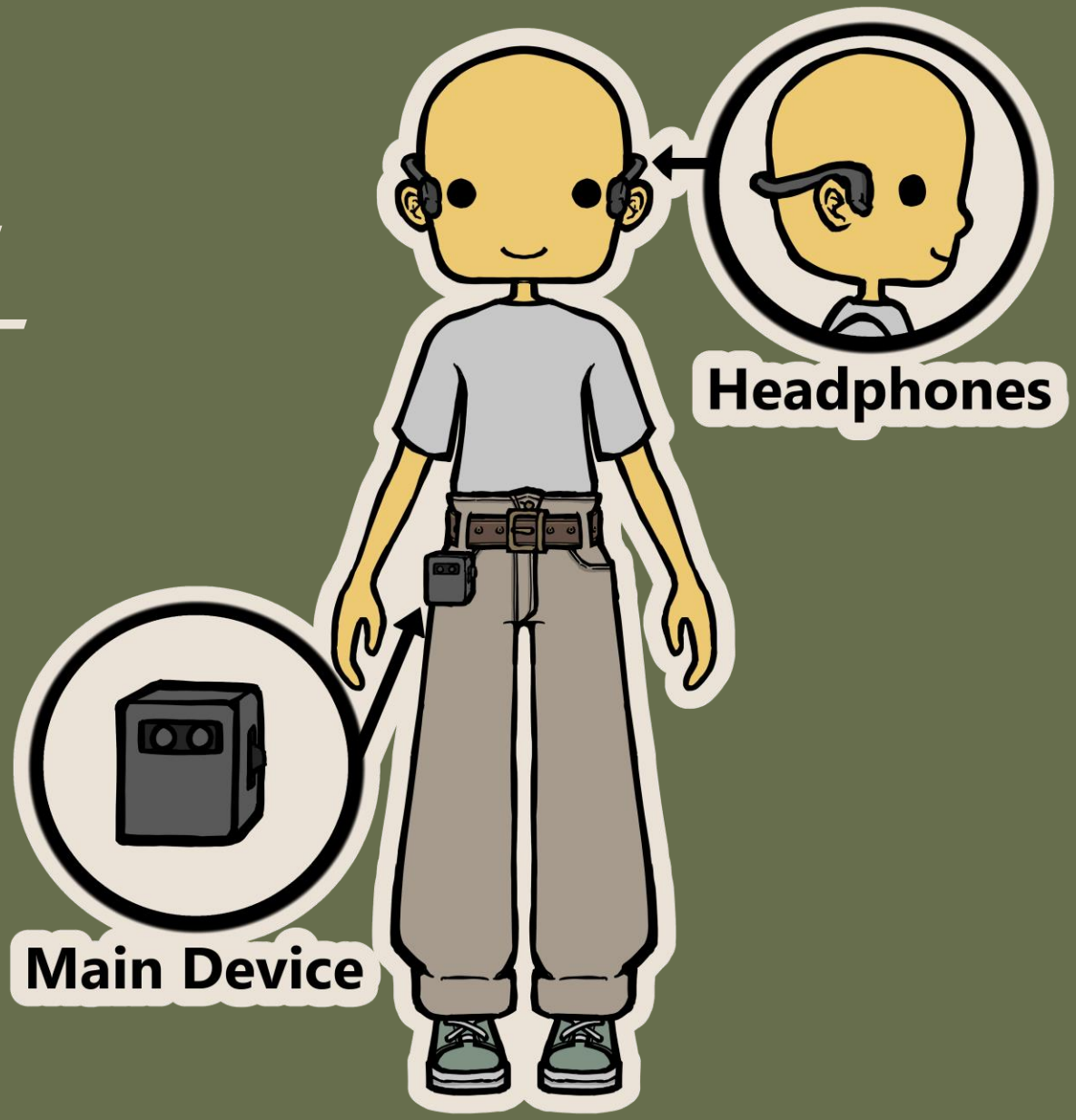


DIAGRAM 2: *OUTER SHELL*

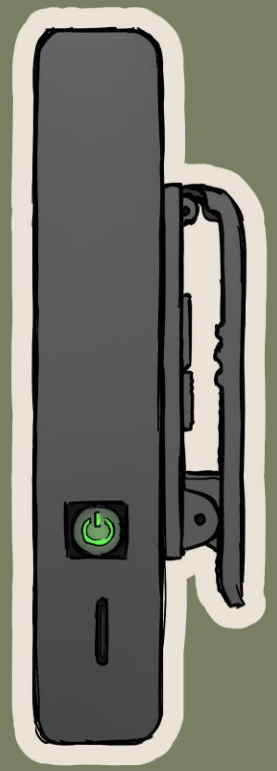
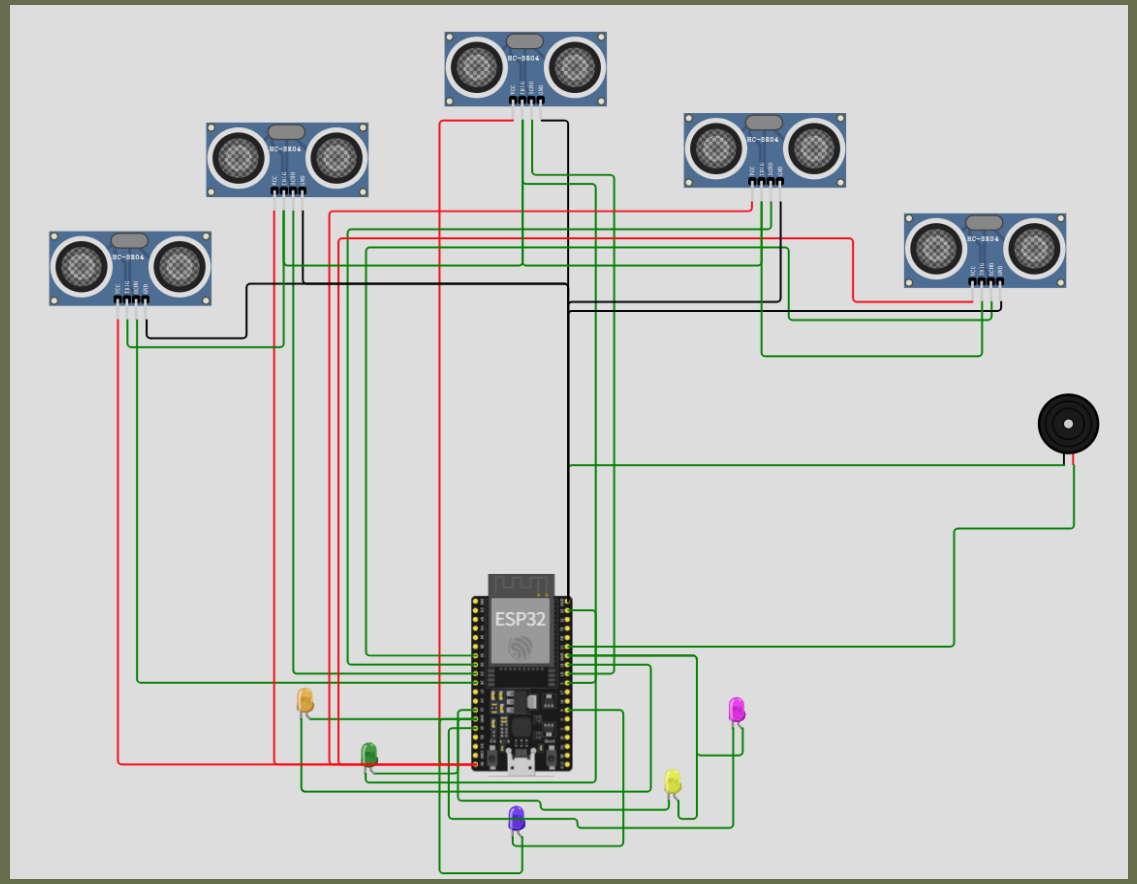
Front	Side	Back
		

DIAGRAM 3: HARDWARE

- Basic Circuit



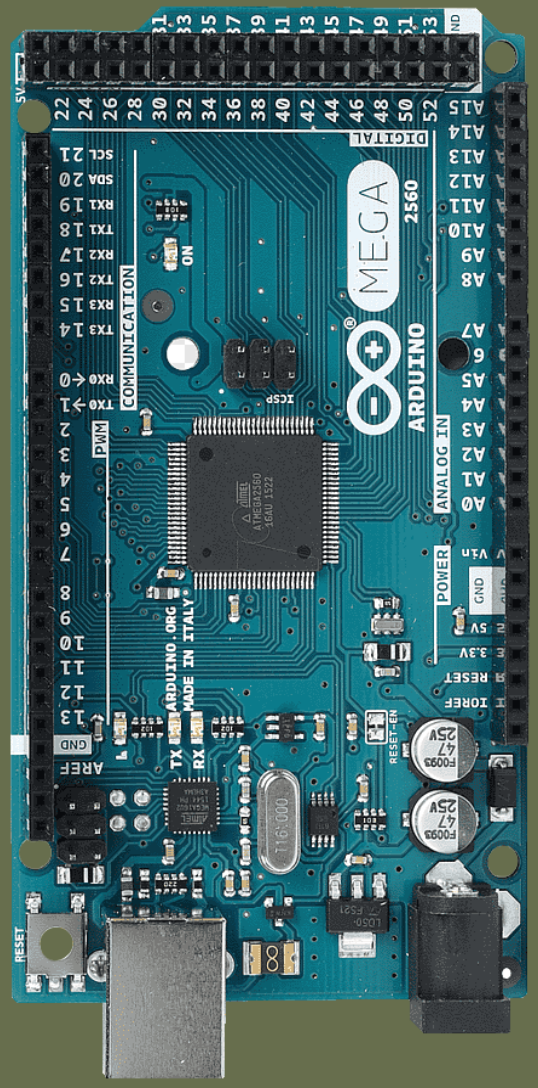
- Principle

The 120 wide-angle scanning is achieved through the five detectors, each of which is paired with a vibrator (LED) and a buzzer. When one of the detectors detects an obstacle, the corresponding vibrator (LED) will start to operate, and at the same time, the buzzer will emit an alarm sound. The closer the sound, the higher the vibration frequency, and the louder the buzzer will be.

https://wokwi.com/projects/475791115642232833

ARDUINO

- Arduino is an open-source platform for electronics projects
- Primary hardware component is a circuit board that houses a microcontroller and pins
- Additional modules can be added as needed
- The microcontroller is the main processor
- A wire will be connected to a pin on one end and another module on the other end. The microcontroller will send an electric current as dictated by its programming



ARDUINO IDE

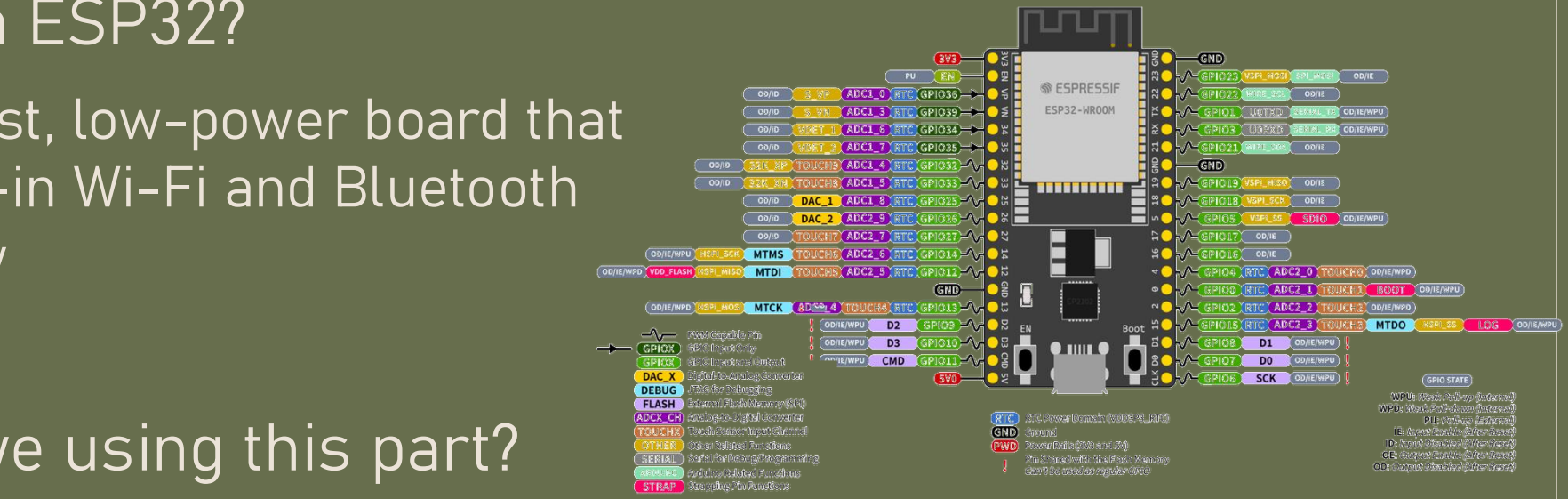
- The microcontroller can be programmed using a simple programming environment called Arduino IDE
- Supported programming languages include C, C++, and assembly
- Arduino IDE is also compatible with other types of microcontrollers such as the ESP32, which we will be using for this project



ESP32

What is an ESP32?

- A low-cost, low-power board that has built-in Wi-Fi and Bluetooth capability



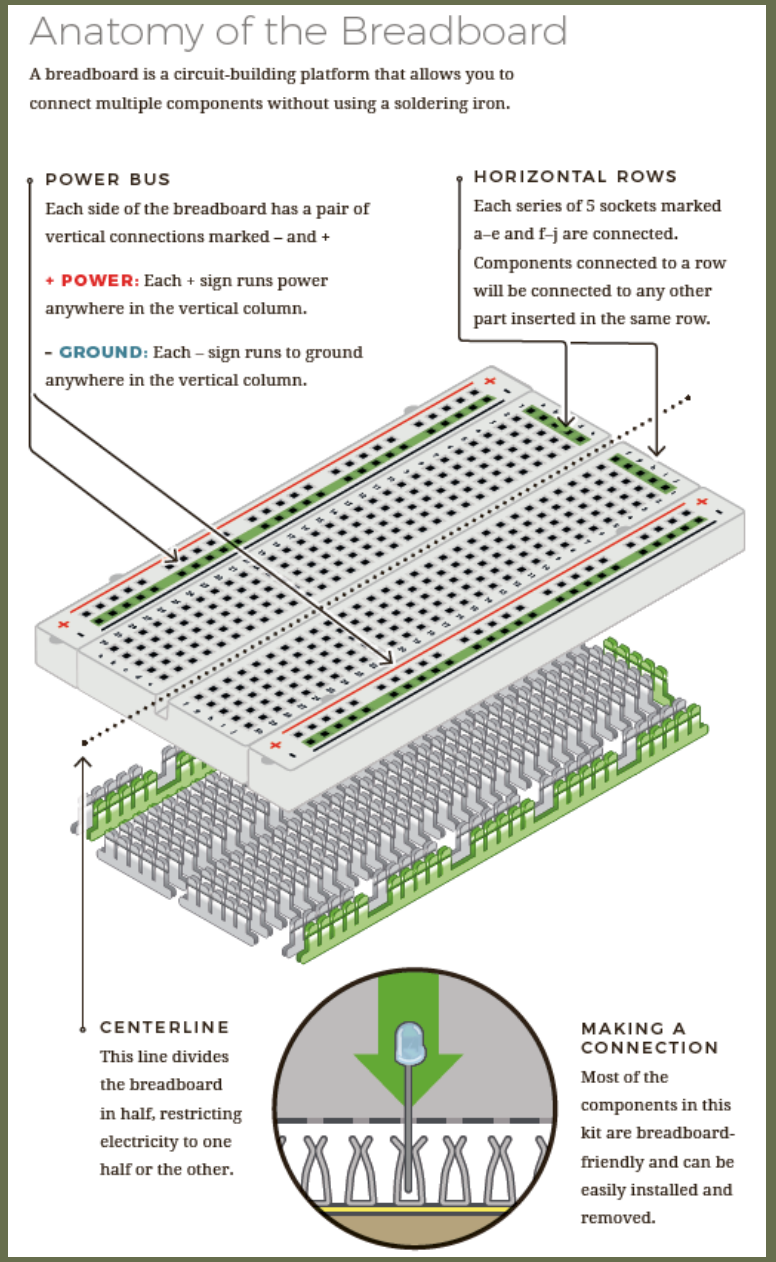
Why are we using this part?

- This will be the brain of our project, it's a microcontroller.
- The ESP32 offers more processing power and memory than entry-level Arduino boards like the Uno, which allows it to handle multiple sensors and more complex logic simultaneously.

BREADBOARD

What is a Breadboard?

- a development board that allows you to connect all of your components without soldering.
- Breadboards have metal lanes inside that allow for currents to flow between components.
- There are special lanes that run perpendicular to the edge of the board called the power bus. They allow you to connect multiple components to the voltage (5V, 3.3V, or VIN) pins and the ground (GND) pins



VOLTAGE/BATTERY



What is Voltage?

- the electrical potential difference that drives current through a circuit, essentially the "push" that moves electricity from the power source to each component.

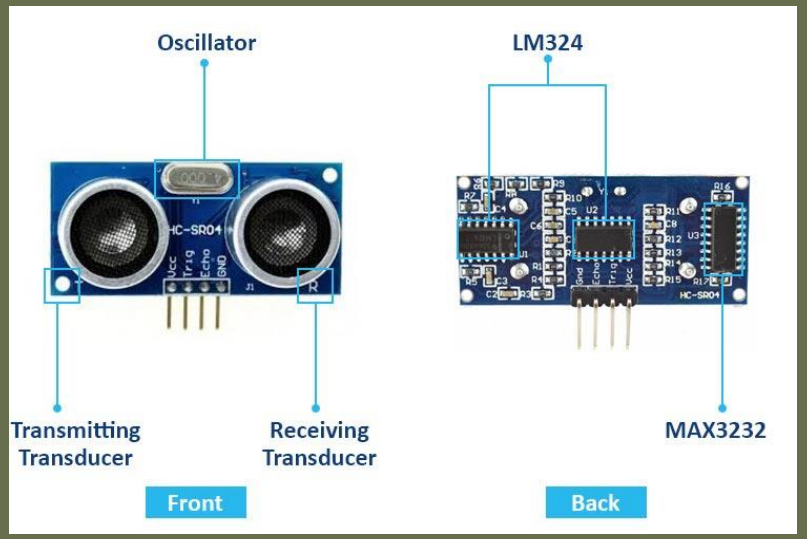
What is important about this?

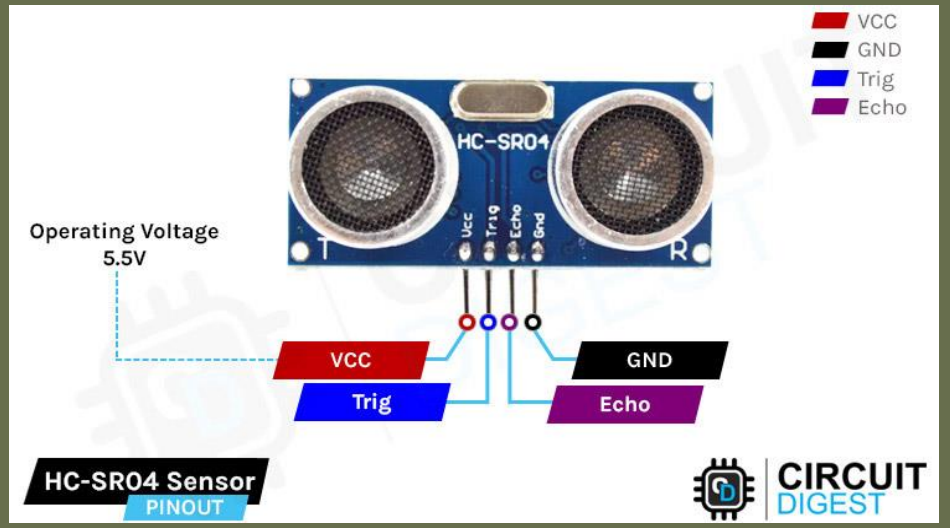
- Our components need to share a compatible voltage (most of our parts run on 3.3V or 5V), which affects how we wire the battery, the ESP32, and each sensor/actuator together.
- We need the right voltage, because a voltage above the limit will cause damage.

ULTRASONIC SENSOR

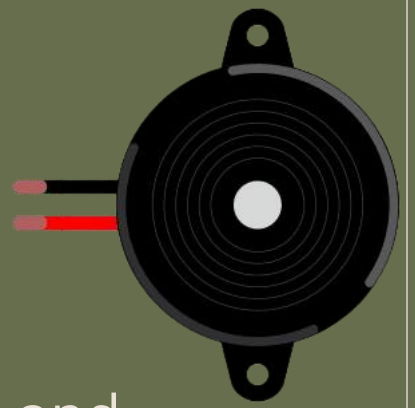
What is an Ultrasonic sensor?

- It's a component that emits a sound pulse, measures time for the echo to return and calculates distance in order to determine how close an object is to the sensor.





BUZZER



- The buzzer module will be one form of audio feedback.
- There are two types of electronic buzzers – active buzzers and passive buzzers
  - Active buzzers only require a DC signal to produce sound. These buzzers can only produce a single fixed sound.
  - Passive buzzers require an AC signal to make sound. Additionally, they can be programmed to make a wide variety of sounds.
- We will be using a passive buzzer because it can be programmed to provide more detailed feedback

VIBRATION MOTOR

- The vibration motor will serve as tactile feedback in addition to the audio feedback from the buzzer.
- There are 3 main types of vibration motor commonly used for haptic feedback:
  - ERM (eccentric rotating mass) motors
  - LRA (linear resonant actuator) motors
  - Piezo actuators
- The ERM motor will be used for this project as it can be easily configured to provide the type of tactile feedback needed.
- It will have to be paired with a transistor to control the voltage.



ERM MOTOR

- ERM Motors function by having an unbalanced weight attached to a mechanical shaft.
- When the motor receives power, it will start spinning, which causes the weight to move in changing directions. This creates a centrifugal force, which then causes vibration.
- The intensity, or amplitude, of the vibration can be directly controlled by changing the voltage.
- Higher frequency = higher amplitude

3D PRINTING: *CAD*

- CAD (Computer-Aided Design) is a kind of software where designers can create, edit, and optimize their designs as either 2D drawings or as 3D models.
  - Parametric design allows for easy changes throughout the design process
  - Precision down to exact measurements
  - Usually Cloud-based, allowing for easy access and collaboration
  - Models can be converted to .stl (Standard Triangle Language) files for 3d printing

BENEFITS OF LEARNING CAD SOFTWARE/AUTODESK FUSION 360

- Free for personal use and for students.
- Used by many professionals outside of computer science, including:



ARCHITECTS

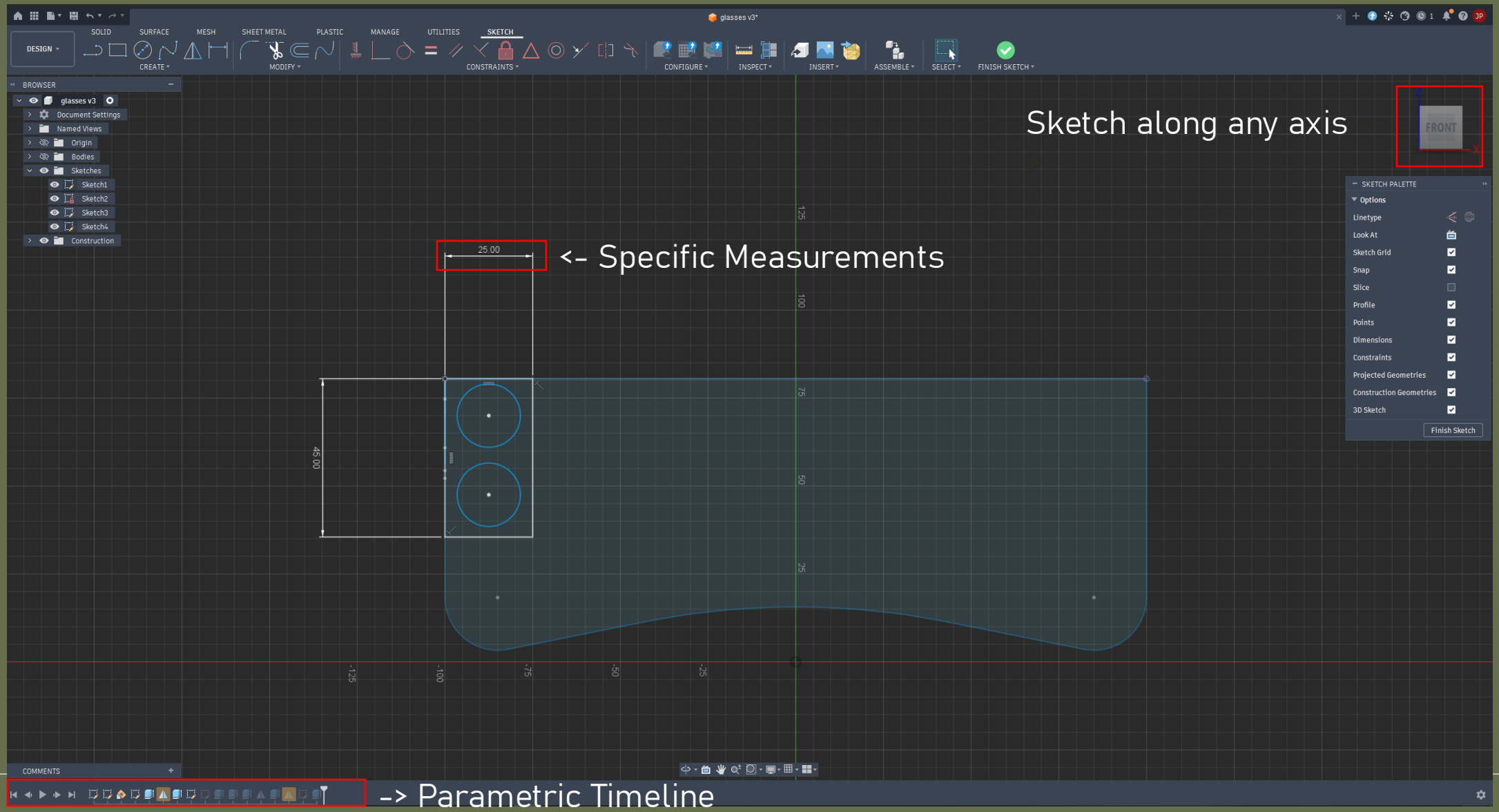


ENGINEERS



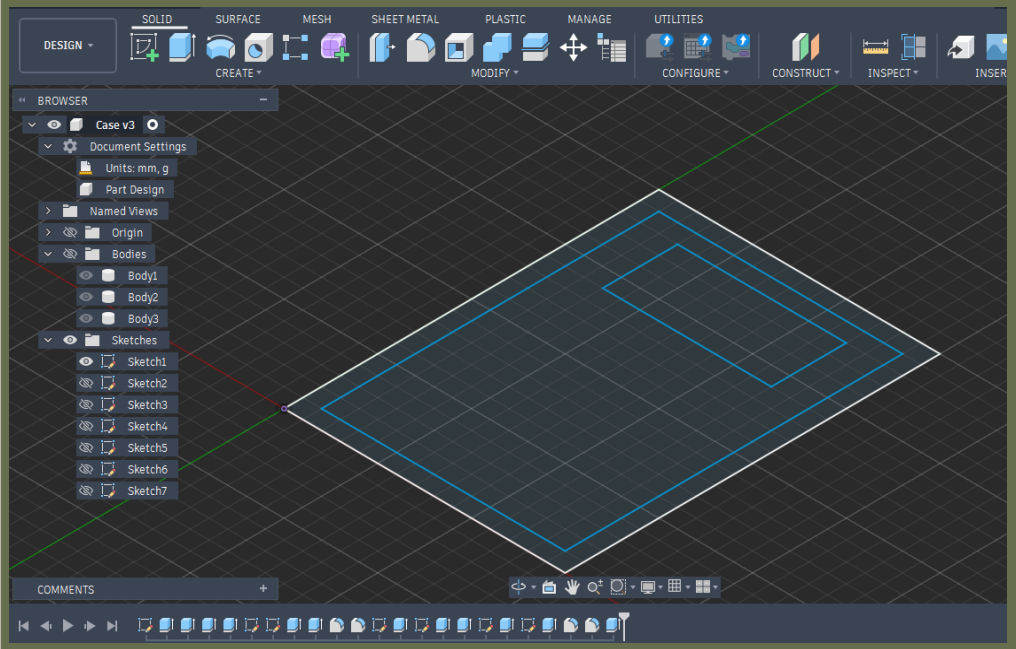
MANUFACTURING

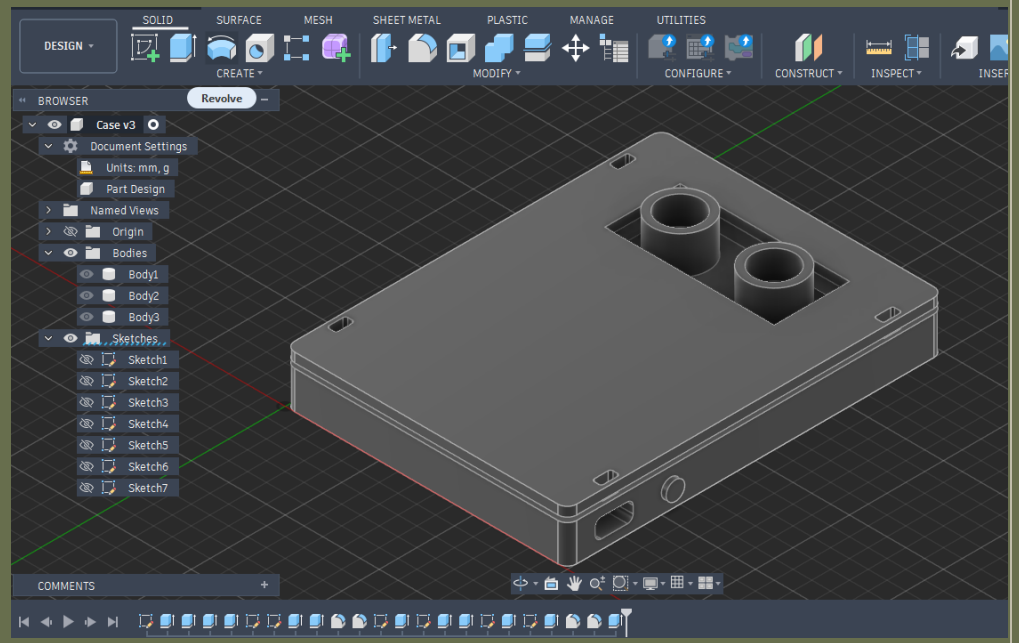
AUTODESK FUSION 360 FEATURES



EARLY CONCEPT RENDERING

*NOT FINAL *



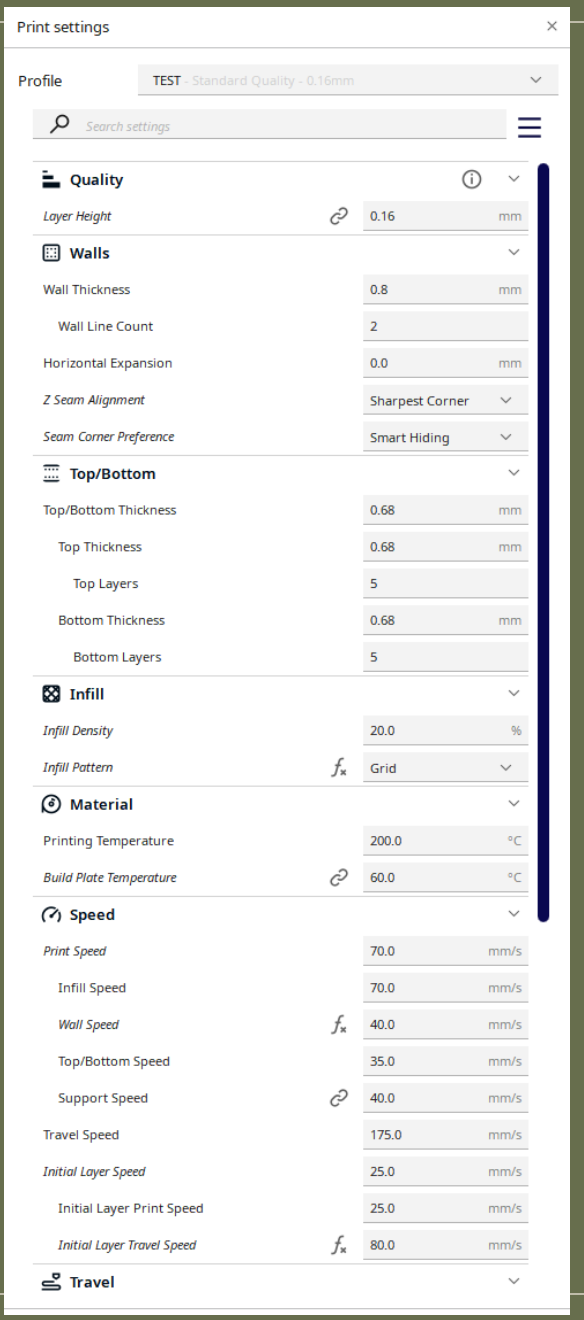


From 2D sketches...

... to a 3D model!

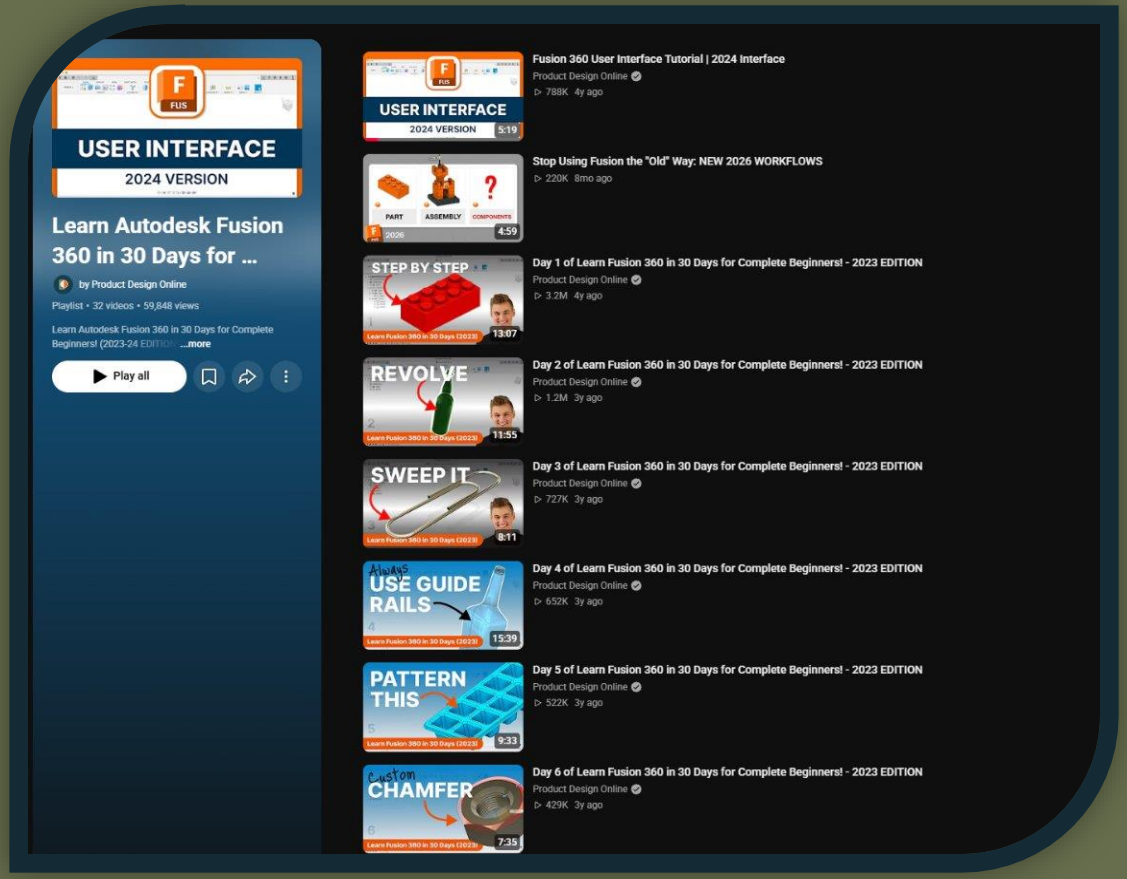
ULTIMAKER CURA/ SLICING SOFTWARE

- A slicing software is a software that converts .stl files into printable G-Code (Geometric code)
- G-Code controls every aspect of the 3d printer, controlling how each layer is printed and with all applied settings.
- Settings include print speed, where to place supports, print temperature, and much more.



GREAT RESOURCE FOR LEARNING AUTODESK FUSION

Learn Autodesk Fusion 360 in 30 Days for Complete Beginners (2024) by Product Design Online



THANK YOU!